

## ASSIGNMENT 4.1

### QUESTION 1

Create a Data Frame (DF) for the given dataset

#### SOLUTION

```
patient_ids <- seq(1, 4)
adm_dates <- c("10/15/2009", "11/1/2009", "10/21/2009", "10/28/2009")
ages <- c(25, 34, 28, 52)
diabetes <- c("Type1", "Type2", "Type1", "Type1")
status <- c("Poor", "Improved", "Excellent", "Poor")

df <- data.frame(patient_id = patient_ids,
  adm_date = as.Date(adm_dates, format="%m/%d/%Y"),
  age = ages,
  diabetes = diabetes,
  status = status)

print(df)
```

#### OUTPUT

|   | patient_id | adm_date   | age | diabetes | status    |
|---|------------|------------|-----|----------|-----------|
| 1 | 1          | 2009-10-15 | 25  | Type1    | Poor      |
| 2 | 2          | 2009-11-01 | 34  | Type2    | Improved  |
| 3 | 3          | 2009-10-21 | 28  | Type1    | Excellent |
| 4 | 4          | 2009-10-28 | 52  | Type1    | Poor      |

### QUESTION 2

Perform the following operation on DF

- Extract PatientID and Age in Subset 1.
- Identify the Type1 patients from DF.
- Count the patient of Poor status.
- Print the summary of the DF
- Find the average age of patient having Diabetes
- Input more patient data from the keyboard.

#### SOLUTION

```
subset1 <- df[, c("patient_id", "age")]
type1_patients <- df[df$diabetes == "Type1", ]
poor_status_count <- sum(df$status == "Poor")
avg_age <- mean(df$age)

print(subset1)
print(type1_patients)
print(poor_status_count)
print(df_summary <- summary(df))
print(avg_age)

new_patient <- data.frame(patient_id = as.integer(readline(prompt="Enter Patient ID: ")),
```

```

    adm_date = as.Date(readline(prompt="Enter Admission Date (MM/DD/YYYY): "),
format="%m/%d/%Y"),
    age = as.integer(readline(prompt="Enter Age: ")),
    diabetes = readline(prompt="Enter Diabetes Type (Type1/Type2): "),
    status = readline(prompt="Enter Status (Poor/Improved/Excellent): ")

```

### OUTPUT

```

patient_id age
1         1  25
2         2  34
3         3  28
4         4  52
patient_id adm_date age diabetes status
1         1 2009-10-15 25    Type1   Poor
3         3 2009-10-21 28    Type1  Excellent
4         4 2009-10-28 52    Type1   Poor

```

[1] 2

```

patient_id      adm_date      age      diabetes
Min.   :1.00   Min.   :2009-10-15   Min.   :25.00   Length:4
1st Qu.:1.75   1st Qu.:2009-10-19   1st Qu.:27.25   Class :character
Median :2.50   Median :2009-10-24   Median :31.00   Mode  :character
Mean   :2.50   Mean   :2009-10-24   Mean   :34.75
3rd Qu.:3.25   3rd Qu.:2009-10-29   3rd Qu.:38.50
Max.   :4.00   Max.   :2009-11-01   Max.   :52.00

status
Length:4
Class :character
Mode  :character

```

[1] 34.75

```

Enter Patient ID: 5
Enter Admission Date (MM/DD/YYYY): 06/04/2005
Enter Age: 12
Enter Diabetes Type (Type1/Type2): Type1
Enter Status (Poor/Improved/Excellent): Excellent

```

### QUESTION 3

Create a list named myList having title "My First List" and criteria having following items:

- Age vector a = (12, 14, 16, 20)
- A two dimensional matrix with 5 rows.
- A score vector with values s = ('First', 'Second', 'Third')

### SOLUTION

```

myList <- list(
  title = "My First List",
  criteria = list(
    age_vector = c(12, 14, 16, 20),
    matrix_2d = matrix(1:10, nrow = 5),

```

```
    score_vector = c('First', 'Second', 'Third')
  )
)

print(myList)
print(myList$criteria)
print(myList$criteria$age_vector)
```

## OUTPUT

```
$title
[1] "My First List"

$criteria
$criteria$age_vector
[1] 12 14 16 20

$criteria$matrix_2d
      [,1] [,2]
[1,]    1    6
[2,]    2    7
[3,]    3    8
[4,]    4    9
[5,]    5   10

$criteria$score_vector
[1] "First" "Second" "Third"

$age_vector
[1] 12 14 16 20

$matrix_2d
      [,1] [,2]
[1,]    1    6
[2,]    2    7
[3,]    3    8
[4,]    4    9
[5,]    5   10

$score_vector
[1] "First" "Second" "Third"

[1] 12 14 16 20
```